Quantification: 877_wt_z081c1-2
==============================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	11
interior lacuna count	7
frame-edge lacuna count	4

Interior lacunae only: mean, median and sample SD (n = 7)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	3515.14	4181	2328.43	px^2
major_axis_length_px	lacuna shape	95.8475	115.634	44.7182	px
minor_axis_length_px	lacuna shape	45.9599	48.0863	19.8876	px
aspect_ratio	lacuna shape	2.051	1.9772	0.4611	unitless
eccentricity	lacuna shape	0.8519	0.8627	0.0716	unitless
solidity	lacuna shape	0.8214	0.8539	0.0911	unitless
roots_count	headline	10.5714	9	6.3208	count
ring_length_r30_px	headline	407	431	203.743	px
ring_length_r60_px	reported	1052.14	1176	451.753	px
owned_length_px	ownership-dependent	2683.81	2777.99	1798.8	px
edge_count	ownership-dependent	129.857	137	80.9577	count
mean_edge_length_px	ownership-dependent	19.2219	19.9571	3.1267	px
perimeter_px	normalised	312.383	354.617	160.701	px
ring_area_r30_px2	normalised	10635.7	11867	3805.27	px^2
ring_area_r60_px2	normalised	25893.7	28242	7282.39	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	1	1	0	unitless
ring_density_r30	normalised	0.036897	0.0363192	0.014095	px^-1
ring_density_r60	normalised	0.039529	0.0442111	0.0134871	px^-1
roots_per_100px_perimeter	normalised	3.4567	3.1019	1.7713	per 100 px
ring_attached_length_r30_px	network v2	358.714	338	199.501	px
ring_attached_length_r60_px	network v2	814.143	911	434.976	px
ring_length_w_r30_px	network v2	465.008	487.517	237.183	px
ring_length_w_r60_px	network v2	1211.33	1347.01	523.369	px
sholl_crossings_r10	network v2	9	9	5.5377	count
sholl_crossings_r20	network v2	12.2857	15	5.5592	count
sholl_crossings_r30	network v2	15.7143	17	7.7398	count
loop_count	loops, report only	11	10	6.5828	count
loop_edge_share	loops, report only	0.3204	0.3359	0.1678	share
ring_width_mean_r30_px	width	7.0973	7.1902	1.119	px

Per field
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measure	value	units
analysed_area_px2	1.0135e+06	px^2
total_skeleton_length_px	32385	px
canalicular_length_density_per_px	0.0319538	px^-1
junction_count	685	count
junction_density_per_px2	0.00067588	px^-2
lacuna_count	11	count

lacunae_per_px2	1.085e-05	px^-2
skeleton_component_count	327	count
mean_component_length_px	99.0367	px
median_component_length_px	25	px
field_length_density_w_per_px	0.0365719	px^-1
field_density_without_flagged_per_px	0.0319538	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	502	px
unattached_component_count	30	count
canalicular_length_density_attached_per_px	0.0319538	px^-1
field_length_density_with_unattached_per_px	0.0324491	px^-1
excluded_area_px2	0	px^2
analysed_field_area_px2	1.04858e+06	px^2
loop_count	99	count
loop_edge_share	0.2461	share
edge_count	1650	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.4376	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.198	px
width_pixels_used	32385	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	828	47.9341	26.2785	1.8241	0.8363	0.811	1.1987	10.5652	544.853	True
2	4064	125.912	44.1728	2.8504	0.9364	0.8472	0.9481	38.4616	829.728	True
3	6500	135.386	68.4723	1.9772	0.8627	0.7101	0.7764	202.316	509.649	False
4	493	36.8679	18.6636	1.9754	0.8624	0.8203	-0.7656	201.503	648.073	False
5	4498	128.704	46.6689	2.7578	0.9319	0.9623	0.3797	397.582	25.7919	True
6	4895	134.554	48.0863	2.7982	0.934	0.8829	0.4584	399.457	314.021	False
7	3078	102.935	62.6793	1.6422	0.7932	0.6805	0.8278	440.301	618.982	False
8	5044	117.23	57.6341	2.034	0.8708	0.8539	1.376	449.833	814.625	False
9	1085	58.3256	24.5501	2.3758	0.9071	0.9313	0.4295	634.654	16.1935	True
10	4181	115.634	46.6872	2.4768	0.9149	0.8959	1.0406	724.949	827.982	False
11	415	28.3253	19.4963	1.4529	0.7254	0.9061	-0.3467	864.431	580.06	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	8	225	625	1132.09	56	20.2158	7.3189
2	8	304	788	1030.69	45	22.9042	7.1897
3	20	720	1581	5700.64	259	22.0102	7.3764
4	7	316	816	1130.15	68	16.6198	8.5032
5	6	247	711	1027.82	44	23.3596	8.1873
6	11	431	1176	2663.89	137	19.4444	8.0356

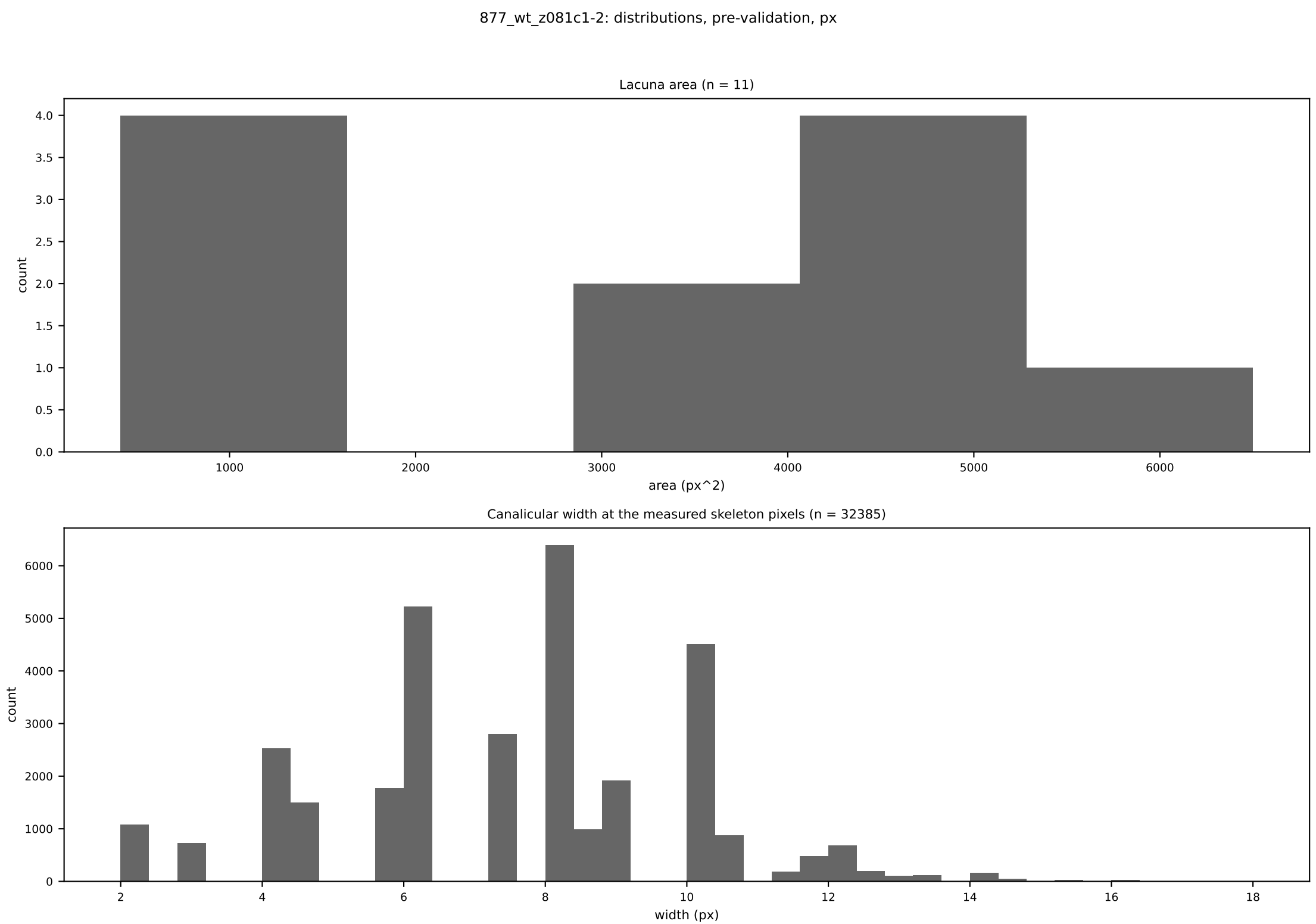
7	9	431	1317	3008.02	141	21.3335	6.7941
8	17	531	1301	3492.49	175	19.9571	7.1902
9	0	65	236	0	0	0	7.9591
10	9	366	980	2777.99	128	21.703	6.7753
11	1	54	194	13.4853	1	13.4853	5.006

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	149.397	4144	11016	0.5976	0.5673	0.0542954	0.0567357	5.3549
2	351.404	9408	21117	0.7903	0.7218	0.0323129	0.0373159	2.2766
3	515.671	15609	34048	1	1	0.0461272	0.0464345	3.8784
4	108.397	5713	15341	1	1	0.0553124	0.0531908	6.4577
5	326.735	7997	18643	0.6576	0.6224	0.0308866	0.0381376	1.8364
6	354.617	12438	30402	1	1	0.0346519	0.0386817	3.1019
7	442.76	11867	27818	1	1	0.0363192	0.0473435	2.0327
8	355.262	12275	29427	1	1	0.0432587	0.0442111	4.7852
9	148.953	4761	12108	0.6788	0.6158	0.0136526	0.0194913	0
10	323.99	11381	28242	1	1	0.0321589	0.0347001	2.7779
11	85.9828	5167	15978	1	1	0.0104509	0.0121417	1.163

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	225	571	264.693	727.34	6	7	7
2	276	705	348.563	910.021	9	8	10
3	710	1495	833.767	1830.83	19	17	25
4	313	780	356.492	933.08	5	9	12
5	205	584	278.966	816.896	6	9	10
6	356	926	487.517	1347.01	10	15	16
7	338	911	490.647	1512.71	6	14	22
8	466	944	611.5	1489.44	12	15	17
9	0	0	74.9411	270.451	0	4	4
10	277	592	416.019	1150.94	9	15	17
11	51	51	59.1127	215.309	2	1	1


877_wt_z081c1-2: distributions, pre-validation, px
Lacuna area (n = 11)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 32385)
count
width (px)